

Ayush Tambde

PROFILE

Programmer with a physics background. Experience in low-level software development, training deep neural networks, and scientific computing. Track record of successful technical projects recognised by BT Young Scientist, Emergent Ventures and Stripe. O-1 visa holder.

EDUCATION

BS Theoretical Physics @ University College Dublin. Oct '23 - May '24 (on leave)

EXPERIENCE

Software Engineer (Rust, TypeScript), San Francisco Compute — June 2024 – Jan 2025

Worked on scheduling systems, API, “Fog”, a **VM orchestrator** for GPU clouds, and **real-time trading engines** for optimal matching of compute.

SELECTED PROJECTS – ON GITHUB

MuZero for International Checkers (Python)

FlashAttention for Apple Silicon, used to train a GPT from scratch on an M1 Mac (**Metal Shading Language** + PyTorch)

Mapped cell signalling pathways in breast cancer cells (both luminal and basal subtypes), and predicted targeted interventions for reprogramming them back to a normal state. (Python)

Compiled programming language for quantum computers, backed by **Unitary Foundation** (C++)

Hybrid symplectic **N-body integrator for simulating interactions between outer solar-system bodies**, **H100-parallelised** version of the Fortran Mercury6 integrator (CUDA)

Hobby OS Kernel with a custom systems programming language (C++)

Quantum computer as an emulated PCI device in QEMU; OS above exposed system calls to send commands to this emulator via DMA (C)

PUBLICATIONS

A Programming Language for Quantum Oracle Construction: <https://arxiv.org/abs/2110.12487>

Cell State Transition Models Stratify Breast Cancer Cell Phenotypes and Reveal New Therapeutic Targets: <https://pubmed.ncbi.nlm.nih.gov/39001416/>

AWARDS

Best Overall Individual @ BT Young Scientist 2023 (BTYS, National High School Science Fair in Ireland)

Stripe Award for Best Software Project @ BTYS '21

Emergent Ventures

Unitary Foundation Research Grant

Ireland's 30u30

Irish Language Debate Team Captain (National Semi-Finalist)

GitHub: <https://github.com/at2005>

Twitter: <https://x.com/atamb>